



INN-RAE: Reversible adversarial examples based on invertible neural networks for facial protection

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ABSTRACT

Reversible adversarial examples can effectively prevent data from being accessed and recognized by unauthorized deep neural network models, but existing methods struggle to balance the visual quality and attack effectiveness of the generated adversarial examples. This paper proposes a method for generating reversible adversarial examples based on invertible neural networks (INN-RAE), achieving an effective unification of high attack success rate and high visual stealthiness. Specifically, during the forward propagation phase of the invertible neural network, both the clean sample and a noise matrix are input simultaneously, and adversarial examples are generated by fine-tuning the noise matrix. When restoring the adversarial examples, the same invertible neural network can be used to achieve high-quality restoration and remove the attack noise, thereby realizing end-to-end reversible adversarial example generation and restoration. Compared with existing reversible adversarial example generation algorithms, INN-RAE achieves state-of-the-art levels of attack success rate on multiple face datasets and face recognition models, while also achieving better visual stealthiness and restoration effects.

1. Introduction

The rapid development of artificial intelligence technology has brought significant improvements in many fields but has also brought new challenges to privacy protection. On social networks, users may upload images containing sensitive personal privacy information, such as selfies, addresses around their homes, and photos with friends. This private information can easily be exploited by unauthorized AI, such as crawling users' selfies for face detection and tracking, obtaining users' identity information for spam and advertising pushes, and even acquiring users' facial information to create illegal face-swapped videos involving pornography and violence. There are currently some methods to protect the privacy of information in images to prevent malicious use by AI. For example, adding adversarial noise to facial images to avoid being used for training face recognition models (Shan et al., 2020), Yuan et al. (2022), Yunqian et al. (2023), Tang et al. (2024), performing style transfer and makeup style changes on facial images to evade detection by face recognition models (Shengshan et al., 2022), adding much random noise to image data to prevent unauthorized personnel from using datasets for model training (Zhao et al., 2022a), Rongke et al. (2024), and adding adversarial noise to facial images to prevent images from being used as deepfake material (Chin-Yuan et al., 2020), Ruiz et al.

(2020), Chin-Yuan et al. (2021), Wang et al. (2022), Aneja et al. (2022). Although the above method can achieve good results in image privacy protection, it adopts a one-way perturbation method, which means that without additional information transmission, the receiver cannot restore the original image, thereby limiting the use of authorized AI. To solve this problem, some scholars have proposed the concept of Reversible Adversarial Examples (RAE), which means that under the premise of pre-communication between the two parties, adversarial examples can be restored so that images can be used legally (Yin et al., 2023), Zhang et al. (2023), Zhen et al. (2024), Liu et al. (2023), Xiong et al. (2023). In summary, the image mentioned above privacy protection methods have the following shortcomings: 1. Protecting datasets requires significant image modifications, resulting in poor visualization and difficulty adapting to social media scenarios. 2. For studies on reversible adversarial examples, RDH techniques can reduce the success rate of attacks, while using generators can degrade the quality of restored images.

This paper proposes a high visual quality Reversible Adversarial Example generation algorithm based on Invertible Neural Networks (INN-RAE) to address the issues above. As shown in Fig. 1, Alice needs to protect her photos. She first trains an INN that can generate and restore adversarial examples, and then distributes it to authorized users. After that, she uses the INN to add noise to her selfie and uploads it to

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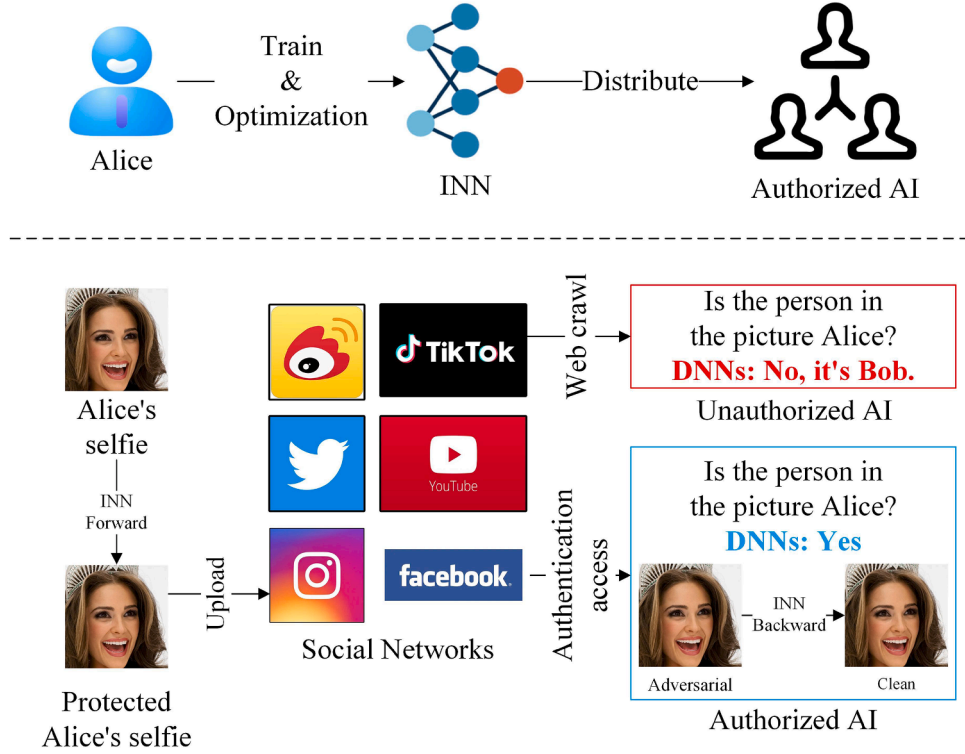


Fig. 1. INN-RAE ensures that images can be used by authorized users and prevents access by unauthorized users.

the public network for sharing. Authorized users can access and analyze the photo normally, while unauthorized users will mistake it for Bob, thus protecting Alice's privacy.

Overall, the contributions of this paper are as follows: 1. This paper proposes a novel reversible adversarial example generation method, INN-RAE, which improves the visual quality and attack success rate of reversible adversarial examples. 2. A new joint optimization framework is proposed, which simultaneously optimizes the model's recognition loss and the visual stealthiness of generated adversarial examples, achieving a balance between the two. 3. Experiments on different face datasets and face recognition models demonstrate that INN-RAE achieves state-of-the-art attack success rates while also attaining better visual stealthiness.

2. Related work

In this section, a brief introduction to reversible adversarial examples and invertible neural network technology will be given.

2.1. Adversarial attack

With the rapid development of artificial intelligence technology based on deep neural networks (DNNs), DNNs have achieved excellent performance in multiple fields. However, the emergence of adversarial examples has raised concerns about the safety of artificial intelligence. A well-performing DNN can be fooled into making completely wrong judgments by adding imperceptible adversarial noise to clean samples. Depending on whether the attacker can access the structure and gradient information of the target model, adversarial attacks can be classified into white-box attacks and black-box attacks. White-box attacks have a higher success rate and better stealthiness against the target model (Goodfellow et al., 2015, Madry et al. (2018), Carlini and Wagner (2017), Dong et al. (2018), Ye et al. (2023), Zhao et al. (2022b)). Black-box attacks are more suitable for adversarial attacks in practical scenarios but are more challenging to execute and leave more apparent traces (Wei et al., 2019, Wei (2022b), Xingxing et al. (2023), Wei

(2022a), Wei et al. (2022), Ran et al. (2025), Rabhi et al. (2024)). The objective of the adversarial attack is shown in Eq. (1), which is to minimize the added perturbation σ on the premise of satisfying the failure of target model $f(\cdot)$ recognition, $|\sigma|_p$ denotes Lp norm.

$$\sigma = \underset{f(X+\sigma) \neq f(X)}{\operatorname{argmin}} |\sigma|_p, X_{adv} = X + \sigma \quad (1)$$

2.2. Reversible adversarial examples

Using adversarial sample technology to add adversarial noise to privacy information, making it impossible for the target network to identify the content correctly. Most adversarial noise addition techniques are one-way, meaning that images with added noise cannot be restored without additional information, limiting the use of these samples in authenticated AI (Shan et al., 2020, Zhao et al. (2022a), Thanh et al. (2023)). To address this issue, researchers have proposed the concept of reversible adversarial samples (Yin et al., 2023), Zhang et al. (2023), Zhen et al. (2024), Liu et al. (2023), Xiong et al. (2023)). The receiver can remove noise from adversarial samples for legitimate use through pre-agreed rules. The generation target is shown in Eq. (2), where X_{RAE} denotes reversible adversarial example, $FA(\cdot)$ denotes forward attack process, X_{re} denotes reconstruction sample, and $BR(\cdot)$ denotes backward reconstruct process.

$$\begin{aligned} X_{RAE} &= FA(f, X), f(X_{RAE}) \neq f(X) \\ X_{re} &= BR(X_{RAE}), f(X_{re}) = f(X) \end{aligned} \quad (2)$$

Although the RAE generation methods mentioned above can restore adversarial examples, RAE based on RDH algorithms can decrease attack success rate and add additional computational overhead. In contrast, RAE based on generators can result in poor-quality generated adversarial images and reconstructed images.

2.3. Invertible neural network

Invertible neural networks have attracted the attention of many researchers due to their highly interpretable network structure (Dinh

et al., 2015), Dinh et al. (2017) and have been deeply applied in many fields, such as image generation (Ardizzone et al., 2019), Ardizzone et al. (2020), image denoising (Jun-Jie & Luigi, 2022), Sutanu and Kumar (2023), image steganography (Shao-Ping et al., 2021), Youmin et al. (2022), and video steganography (Chong et al., 2023). Additionally, some studies have utilized INNs to generate adversarial samples. For instance, AdvINN (Chen et al., 2023) employs INN to optimize perturbations in clean samples, resulting in highly concealed adversarial noise. However, this method does not consider the issue of adversarial sample reconstruction, leading to problems such as low reconstruction quality and difficulty correctly identifying the reversed reconstructed images.

3. Reversible adversarial examples based on invertible neural network

This section will model the privacy protection scenario of reversible adversarial examples on social networks and determine the optimization objectives. Then, we will introduce the INN-RAE proposed in this paper in detail.

3.1. Problem formulation

We hereby provide a definition for the scenario of invertible adversarial examples. For unauthorized AI, the protected images should cause misclassification in their DNNs. In contrast, the original images should be recoverable for authorized AI, and the content should be correctly recognized using legal DNNs. Therefore, the optimization goals in this scenario are as shown in Eq. (3).

$$\begin{aligned} \min D(X_{adv}, X) + D(X_{re}, X) \\ \text{s.t. } f(X_{adv}) \neq f(X) \text{ and} \\ f(X_{re}) = f(X) \end{aligned} \quad (3)$$

For a given original image X and target network f , generate a protected image X_{adv} such that the difference between X_{adv} and X is minimized, $D(\cdot)$ denotes image distance measurement function. However, the target network simultaneously misidentifies X_{adv} . For the recovered image X_{re} , the difference between X_{re} and X should be minimized, and the recognition results of the target network for X and X_{re} should be consistent. Integrating the optimization as mentioned above goals, we obtain

a unified optimization objective function as shown in Eq. (4).

$$\begin{aligned} X_{adv}, X_{re} = \operatorname{argmin} D(X_{adv}, X) + D(X_{re}, X) \\ + D(f(X_{re}) \text{ allel } f(X)) - D(f(X_{adv}) \text{ allel } f(X)) \end{aligned} \quad (4)$$

In the subsequent sections, this paper will analyze and optimize privacy protection in image classification and face recognition scenarios based on the above-mentioned optimized objective function.

3.2. Forward attack & backward reconstruction

Fig. 2 shows the overall flowchart of the reversible adversarial sample generation algorithm based on invertible neural networks proposed in this paper. The entire process can be divided into two steps. The first step is the forward attack by the image owner to generate adversarial noise for the protected image. The second step is the reverse restoration and reconstruction process performed by the legitimate receiver of the image, which restores the reversible adversarial sample to a normally recognizable sample. For an input clean sample X , an adversarial sample X_{adv} is first generated through a forward attack process, and the target network f is queried using the adversarial sample.

$$\text{let } PM = X, X_{adv} = INN_{forward}(X, PM) \quad (5)$$

Then, the generated adversarial sample is passed through the INN's reverse reconstruction process to obtain the reconstructed sample X_{re} .

$$X_{re} = INN_{backward}(X_{adv}) \quad (6)$$

In the forward attack phase, the image $X \in R^{C \times H \times W}$ that needs to be protected is also used as the initial perturbation matrix PM , which adds perturbations for updating. DWT transformations are performed to obtain frequency domain information $X^0 \in R^{4C \times H/2 \times W/2}$ and $PM^0 \in R^{4C \times H/2 \times W/2}$. At this time, the image resolution of the frequency domain information is one-quarter of the original image, which facilitates the training of the INN. Afterwards, X^0 and PM^0 are put into the INN for forward deduction. The INN consists of N invertible blocks, and the structure of each invertible block is shown in Fig. 3. The K -th invertible block's input is X^k and PM^k , and the output is X^{k+1} and PM^{k+1} , where X^k and X^{k+1} have the same shape, and PM^k and PM^{k+1} have the same shape. The forward calculation formula of the invertible block k is shown as Eqs. (7) and (8). Where ρ'_k and μ'_k represent specific feature extractors, the five-layer dense block (Gao et al., 2017) is used for

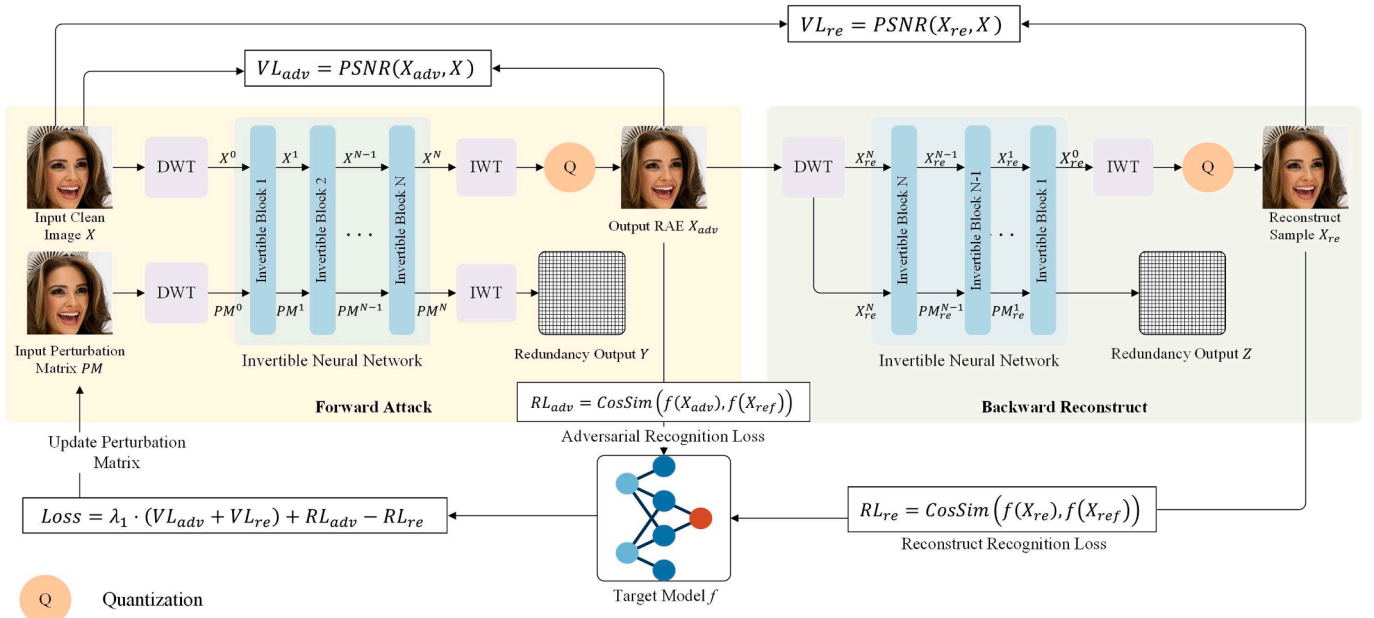


Fig. 2. Schematic diagram of the overall process of INN-RAE.

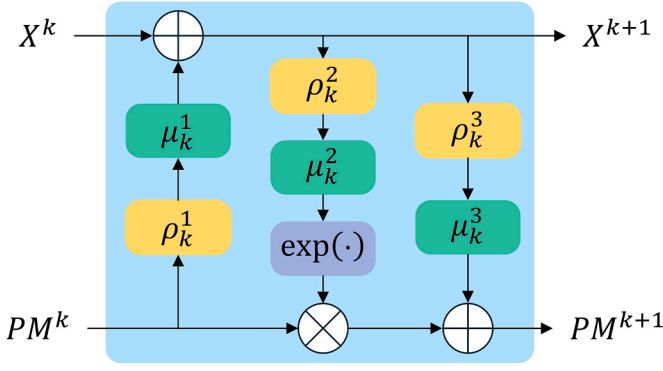


Fig. 3. Illustration of invertible block k used in INN, where $\otimes$ denotes element-wise multiplication, $\oplus$ denotes element-wise addition.

feature extraction. Then, a 1×1 convolution is connected to compress the channels of the output image, ensuring the consistency of the input and output shapes.

$$X^{k+1} = X^k \oplus \mu_k^1(\rho_k^1(PM^k)) \quad (7)$$

$$PM^{k+1} = PM^k \otimes \exp(\mu_k^2(\rho_k^2(X^{k+1}))) \oplus \mu_k^3(\rho_k^3(X^{k+1})) \quad (8)$$

After the forward pass through N invertible blocks, the frequency domain output X^N of the adversarial sample is obtained through quantization function Q :

$$X_{adv} = Q(IWT(X^N)) \quad (9)$$

where quantization function Q is shown in Eq. (10).

$$Q(X) = T(R(X \times 255)/255) \quad (10)$$

Where $R(\cdot)$ denotes the rounding function, $T(\cdot)$ represents the truncation operation, which truncates the input to within the range of $[0,1]$ to ensure that the output sample forms a usable sample. In addition, the output at the PM end will result in a redundant output Y . When selecting the initial value for PM , an image identical to the original clean sample is chosen to ensure the visual quality of the generated adversarial sample. The receiver's input is a single adversarial example X_{adv} during the backward reconstruction phase. After applying the DWT transform to the X_{adv} , it is used as a dual-channel input X_{re}^N and fed into the invertible blocks of the INN in reverse order. For the k -th block, the backward computation is shown in Eqs. (11) and (12). Since the computation in the INN is fully reversible, the same network and parameters used in the forward attack are utilized.

$$PM_{re}^{k-1} = (PM_{re}^k \ominus \mu_k^3(\rho_k^3(X_{re}^k))) \otimes \exp(-\mu_k^2(\rho_k^2(X_{re}^k))) \quad (11)$$

$$X_{re}^{k-1} = X_{re}^k \ominus \mu_k^1(\rho_k^1(PM_{re}^k)) \quad (12)$$

After passing through the inverse INN output, the frequency domain sample X_{re}^0 of the reconstructed sample is obtained, and it becomes the reconstructed sample:

$$X_{re} = Q(IWT(X_{re}^0)) \quad (13)$$

3.3. INN-RAE optimization generation

When generating adversarial samples X_{adv} , the parameters and structure of the INN will not be changed, ensuring that the sender and receiver can correctly generate and restore clean samples after obtaining the same INN. When generating adversarial samples, the loss function can be divided into the visual quality loss function and the model classification loss function. The visual quality loss function aims to make the generated adversarial samples and the reconstructed samples have a visual effect as close to the original image. Therefore, we use the peak signal-to-noise ratio (PSNR) function to measure the visual effect. Normalize two images X_1 and X_2 , of the same size to the range of $[0,1]$, with the number of pixels being $n = C \times H \times W$, where C, H, W denotes

images' channel, height, and width. X_i represents the value of the i -th pixel. Calculate the Mean Squared Error (MSE) between the two images and compute their PSNR value via Eq. (15). A more enormous PSNR value means better image visual quality.

$$MSE(X_1, X_2) = \frac{1}{n^2} \sum_{i=1}^n (X_{1i} - X_{2i})^2 \quad (14)$$

$$PSNR(X_1, X_2) = 20 \cdot \log_{10} \left(\frac{1}{MSE(X_1, X_2)} \right) \quad (15)$$

Regarding facial privacy protection, assuming user Alice hopes that unauthorized AI cannot correctly identify and track the photos she posts on social networks. Here, we assume that the unauthorized AI has one or more confirmed photos of Alice herself, denoted as $X_{Alice-ref}$. When Alice posts a new photo X , the thief will compare photo X with the existing reference photos. If the similarity between the two reaches a threshold th , X is assumed to be a photo of Alice. Otherwise, it cannot be identified. Furthermore, regarding targeted attacks on face recognition models, Alice hopes that the unauthorized AI identifies X as Bob, meaning that the similarity between X and Bob's reference image $X_{Bob-ref}$ is greater than the threshold. The similarity between X and Alice's reference image $X_{Alice-ref}$ is less than the threshold th . For a face recognition model f , the input is a face image, and the output is the corresponding one-dim feature vector. The similarity between two face images can be obtained by comparing the distances between the feature vectors of different face images. For an untargeted attack scenario, given a protected clean sample X , a reference image X_{ref} held by an unauthorized AI, and the generated adversarial face image X_{adv} , there are visualization loss VL_{adv} and face recognition loss RL_{adv} , with calculation formulas shown in Eq. (16) and (17). $CosSim(\cdot)$ denotes the cosine similarity function.

$$VL_{adv} = PSNR(X_{adv}, X) \quad (16)$$

$$RL_{adv} = CosSim(f(X_{adv}), f(X_{ref})) \quad (17)$$

For targeted attack scenarios, given a target identity face image X_{t-ref} , the corresponding targeted attack recognition loss is RL_{adv} , and its calculation method is shown in Eq. (18).

$$RL_{adv} = -CosSim(f(X_{adv}), f(X_{t-ref})) \quad (18)$$

Similarly, for X_{re} obtained through reverse reconstruction using INN, to ensure that authorized AI can correctly identify its identity, the reconstruction recognition loss RL_{re} is constructed, and its calculation method is shown in Eq. (19).

$$RL_{re} = CosSim(f(X_{re}), f(X_{ref})) \quad (19)$$

Finally, when calculating the unified loss function $L(f, X, PM)$, the visual loss and recognition loss are comprehensively considered to obtain the final optimization objective function, which is given in Eq. (20).

$$L(f, X, PM) = \lambda_1 \cdot (VL_{adv} + VL_{re}) + RL_{adv} - RL_{re} \quad (20)$$

The final optimization iteration algorithm is shown in Eqs. (21) and (22), and the adversarial sample X_{adv} and reconstructed sample X_{re} that meet the requirements are obtained by continuously updating PM .

$$g_{m-1} = \nabla_{PM_{m-1}} L(f, X, PM_{m-1}) \quad (21)$$

$$PM_m = PM_{m-1} + \alpha \cdot \text{sign}(g_{m-1})$$

During the updating process of PM and X_{adv} , L_∞ norm to constrain both, and for ease of control, both adopt the same attack magnitude for limitation. ϵ is the maximum attack amplitude.

$$\begin{aligned} |X_{adv} - X|_\infty &\leq \epsilon \\ |PM - X|_\infty &\leq \epsilon \end{aligned} \quad (22)$$

4. Experiments and analysis

4.1. Experiments setting

For the face recognition task, the VggFace2 (Qiong et al., 2018) and CelebA (Zhang et al., 2020) datasets were used for evaluation. VggFace2

Table 1
Experimental results of untargeted attacks on CelebA, with the best results in **bold**.

TargetModels	SRAE	RAE-FGSM (post)	RAE-BIM (post)	RAE-BIM (in-the-loop)	INN-RAE (Ours)
CelebA					
ArcFace	0.8816	0.8866	0.9408	0.9600	0.9599
CosFace	0.6681	0.9163	0.9350	0.9504	0.9665
MobileFacenet	0.7898	0.9002	0.9079	0.9526	0.9905
Sphere Face	0.7885	0.9035	0.9306	0.9493	0.9815
Sphere Face2	0.7518	0.9169	0.9271	0.9294	0.9656
Sphere FaceR	0.5665	0.8974	0.9084	0.9642	0.9507
VggFace2					
ArcFace	0.5481	0.9061	0.9086	0.9851	0.9737
CosFace	0.4773	0.9790	0.9471	0.9265	0.9653
MobileFacenet	0.6309	0.9230	0.9902	0.9181	0.9999
Sphere Face	0.7098	0.8813	0.9323	0.9294	0.9902
Sphere Face2	0.8846	0.9048	0.9043	0.9581	0.9622
Sphere FaceR	0.8345	0.9070	0.8992	0.9878	0.9850

contains 3.31 million images of 9131 faces from different people. The CelebA dataset contains over 200,000 facial images from 10,177 celebrities. We randomly selected 1000 faces from 100 individuals, with ten images for each ID, one of which was used as a reference image and the remaining nine as images to be attacked. All the face image sizes are 112×112.

4.1.1. Target model

We selected CosFace (Hao et al., 2018), ArcFace (Jiankang et al., 2019), Sphere Face (Weiyang et al., 2017), and MobileFaceNet (Chen et al., 2018) networks for the face recognition task. These networks have all been trained on VGGFace2 and achieved the accuracy claimed in their original papers.

4.1.2. Attack parameters

The maximum attack amplitude in RAE (Liu et al., 2023) and INN-RAE is designed to be 4/255 to ensure stealthiness. However, since SRAE (Zhang et al., 2023) uses a generator to generate images, setting the attack amplitude too small will result in a low attack success rate, making it impossible to attack successfully. Therefore, its attack amplitude is set to 1, the same as in the original paper, and on this basis, we try to optimize the perturbation amplitude.

4.1.3. Metrics

In experiments, the metrics for evaluating RAE can be divided into visual quality and attack effectiveness. In terms of visual quality metrics, it is necessary to measure the visual quality of the generated adversarial examples and the visual quality of the reconstructed examples. PSNR, SSIM, and L2 norm are metrics to measure visual quality. Here, subscripts distinguish between the visual quality metrics of adversarial and reconstructed samples. When evaluating face privacy protection scenarios, besides general image quality evaluation indicators, we introduced the EQBM (Ou et al., 2023) indicator to measure the quality of face images in the generated adversarial samples. In terms of attack effectiveness metrics, the attack success rate (ASR) of adversarial examples and the cognition error rate (CER) of the reconstructed samples are used for evaluation. A higher attack success rate indicates better generated adversarial examples, while a lower cognition error rate indicates better usability of the recovered samples.

4.2. Attack performance evaluation

When comparing attack effectiveness, we selected two state-of-the-art reversible adversarial example generation algorithms for comparison: SRAE (Zhang et al., 2023) and RAE (Liu et al., 2023). For the RAE method, we chose RAE-FGSM (post), RAE-BIM (post), and RAE-BIM (in-the-loop) for comparison. The experimental results for non-targeted attacks on the CelebA and VggFace2 datasets are shown in Table 1. It

Table 2

For the attack success rate of **targeted attacks** on face recognition models, the best results are highlighted in **bold**.

Target Models	RAE-FGSM (post)	RAE-BIM (post)	RAE-BIM (in-the-loop)	INN-RAE (Ours)
CelebA				
ArcFace	0.2111	0.6751	0.8022	0.8523
CosFace	0.2804	0.4955	0.4975	0.9176
Mobile-Facenet	0.2111	0.7617	0.7772	0.9112
Sphere Face	0.1480	0.7735	0.8972	0.9076
Sphere Face2	0.1840	0.5507	0.8092	0.8940
Sphere FaceR	0.1400	0.4467	0.5291	0.8836
VggFace2				
ArcFace	0.5682	0.9360	0.9879	1.0000
CosFace	0.3200	0.8899	0.9511	0.9842
Mobile-Facenet	0.5682	0.9437	0.9731	1.0000
Sphere Face	0.2626	0.6538	0.7468	0.7742
Sphere Face2	0.2323	0.5361	0.5748	0.5968
Sphere FaceR	0.3131	0.6106	0.6647	0.7364

can be observed that INN-RAE achieved the best attack success rate in most scenarios, maintaining an attack success rate above 95%. In some models and datasets where it did not achieve the optimal attack success rate, the difference between INN-RAE and the most advanced methods was also around 1%. Therefore, it can be considered that INN-RAE has reached the SOTA level in reversible adversarial examples for non-targeted facial attacks.

For the experimental results of targeted attacks, as shown in Table 2, due to the difficulty in achieving targeted facial attacks on human faces using the GAN network-based SRAE method, only the targeted attacks of the RAE method are compared in this paper. As can be seen from the table, INN-RAE achieved the best targeted attack success rate across all tested models and datasets. Particularly, when attacking ArcFace and MobileFacenet in VggFace2, it reached a 100% attack success rate. In addition, we found that the RAE-BIM (post) and RAE-BIM (in-the-loop) methods exhibited significant differences in attack effectiveness across different datasets for targeted attacks on CosFace. In CelebA, they achieved a success rate of less than 50%, but in VggFace2, their success rates were 88.99% and 95.11%, respectively. We speculate that this is due to the significant differences in data distribution between the CelebA and VggFace2 datasets, leading to large variations in effectiveness when attacking the CosFace model.

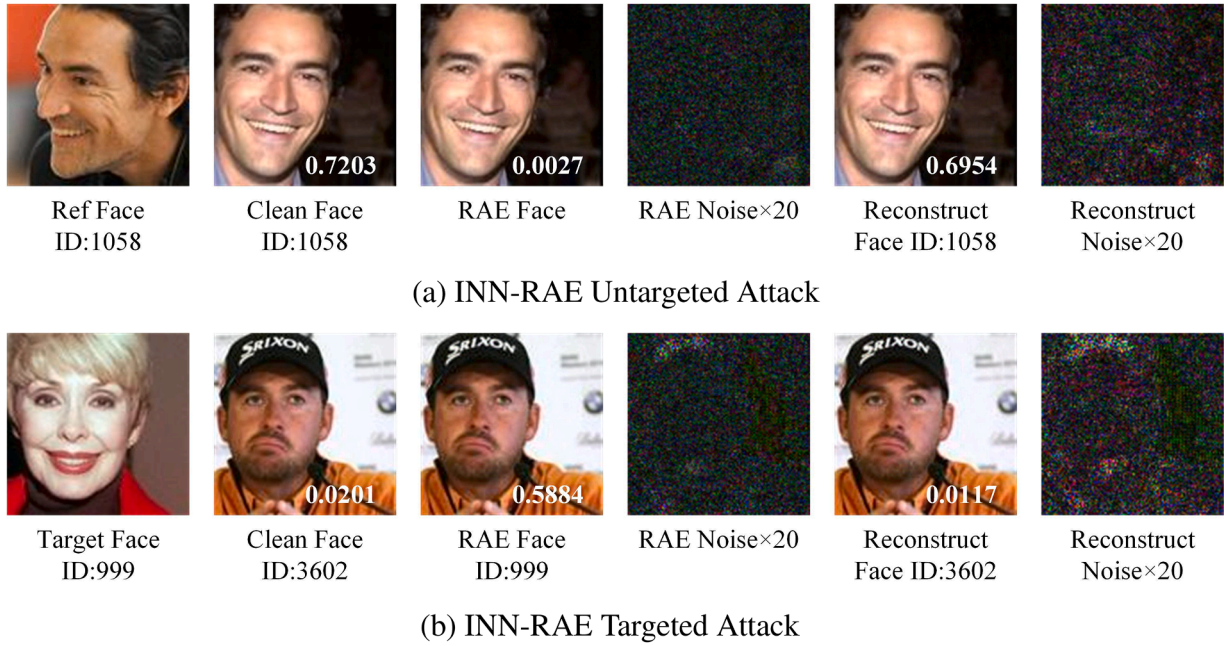


Fig. 4. The untargeted and targeted adversarial attack samples and their corresponding reconstructed samples generated by the ArcFace model on the CelebA dataset by INN-RAE.

Examples of adversarial samples and reconstructed samples generated by this attack method for non-targeted and targeted attacks on the CosFace model using the CelebA dataset are shown in Fig. 4. The similarity between the sample and the reference image is indicated in the bottom right corner of each image. A similarity above the threshold is considered to indicate the same face, while a similarity below the threshold is considered to indicate a different face.

4.3. Attack stealthiness evaluation

When measuring the stealthiness of different reversible adversarial sample generation algorithms, we selected four metrics for evaluation on the CelebA dataset: PSNR, L2, SSIM, and EQBM. The first three are used to objectively measure the degree of image alteration, while the fourth assesses the naturalness of the image. The stealthiness metrics of adversarial attacks generated for different models on the CelebA dataset are shown in Fig. 6. The horizontal axis of each figure represents different target models, while the vertical axis represents the corresponding metric values. It can be observed from the figures that INN-RAE achieves the best visual stealthiness across most models. Only in the EQBM metric, the adversarial samples for the CosFace model are slightly lower than a few other methods, such as RAE-BIM (post). These results demonstrate that INN-RAE has outstanding performance in visual stealthiness (Fig. 5).

Adversarial samples generated using different reversible adversarial attack algorithms and their corresponding perturbations are shown in Figs. 7 and 8. For ease of observation, the generated noise is multiplied

by 20. It is evident that the noise generated using the SRAE method is more pronounced, and the quality of the original image is worse. The noise generated by the RAE method exhibits a noticeable super-pixel upsampling effect, which has the advantage of reducing the required steganographic capacity but the disadvantage of lowering the attack success rate. The adversarial noise generated by INN-RAE is more concealed, with subtle differences from the original image, barely affecting normal usage.

4.4. Reversibility evaluation

When evaluating the reversibility metrics of reversible adversarial samples, we assess them from two perspectives: the similarity between the recovered image and the original image, and the proportion of correctly recognized recovered images. The former is evaluated using PSNR, SSIM, L2 distance, and EQBM, while the latter is assessed using CER. The reversibility test results on the CelebA dataset are shown in Table 3.

As can be seen from the table, compared to the original SRAE, the proposed SRAE in this paper has a lower recognition error rate after recovery and a smaller difference between the recovered image and the original image, demonstrating the excellent performance of our method in terms of reversibility. It is worth noting that the RAE method utilizes reversible data hiding techniques for reversible recovery of adversarial noise, theoretically achieving lossless noise hiding and recovery. Therefore, our method is not compared with RAE in terms of reversibility.

Table 3

Comparison of reversible adversarial sample restoration capabilities, with the best results highlighted in bold.

Target Models	CER ↓		PSNR ↑		SSIM ↑		L2 ↓		EQBM ↑	
	INN-RAE	SRAE	INN-RAE	SRAE	INN-RAE	SRAE	INN-RAE	SRAE	INN-RAE	SRAE
ArcFace	0.0097	0.0968	40.4570	37.3552	0.9801	0.9593	1.8444	2.6319	0.4684	0.4072
CosFace	0.0130	0.3810	40.3975	37.4096	0.9807	0.9575	1.7216	2.6152	0.4615	0.4109
MobileFacenet	0.0136	0.0968	40.4771	37.3552	0.9802	0.9593	1.8399	2.6319	0.4385	0.4072
Sphere Face	0.0150	0.1932	40.3548	37.3605	0.9821	0.9607	1.8673	2.6302	0.4471	0.4187
Sphere Face2	0.0064	0.1290	40.0411	37.1324	0.9820	0.9568	1.9343	2.7022	0.4409	0.4233
Sphere FaceR	0.0147	0.2091	40.0803	37.5167	0.9817	0.9620	1.9255	2.5867	0.4463	0.4407

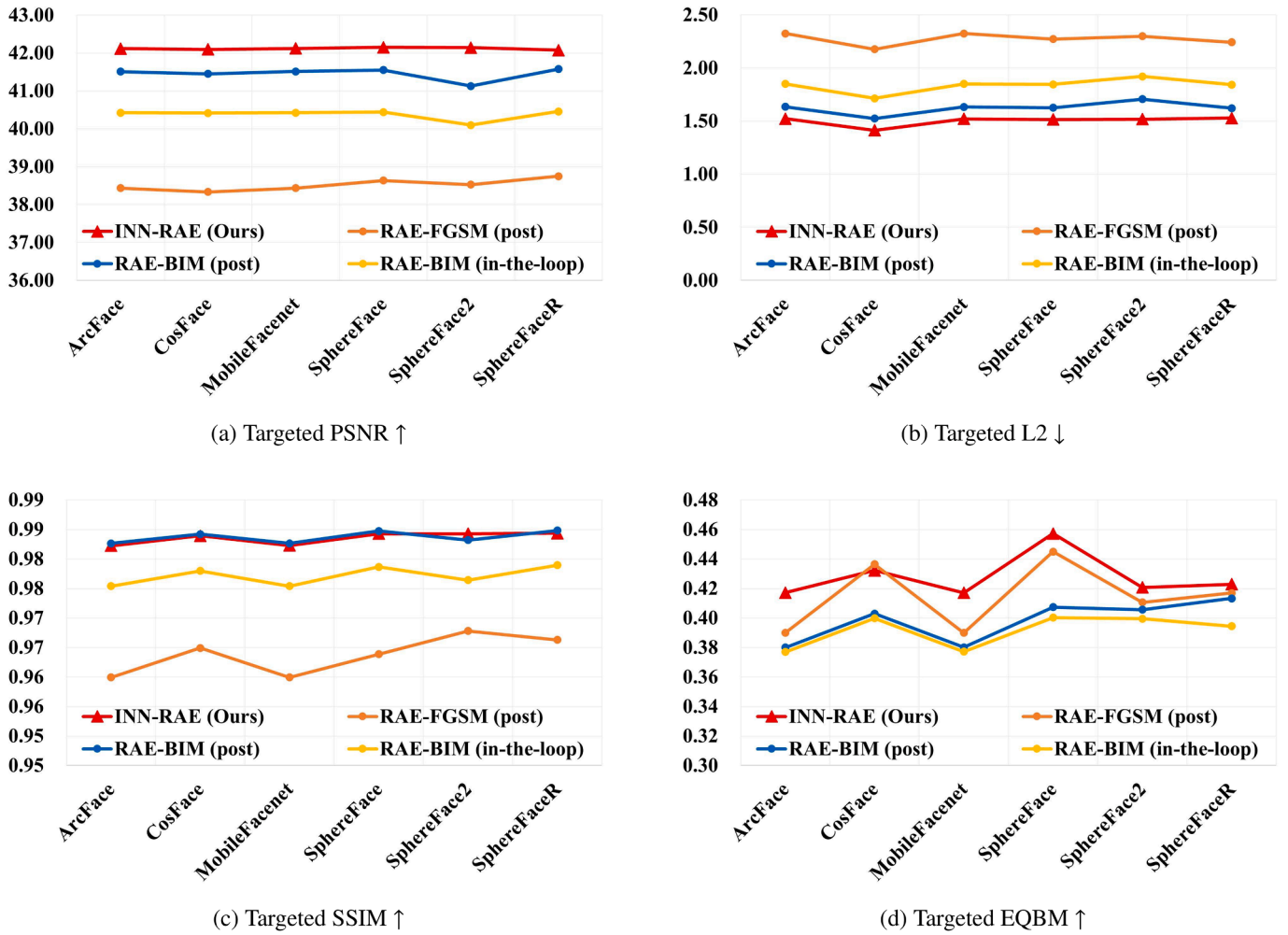


Fig. 5. Targeted adversarial attack stealthiness performance. Where the x-axis represents the target model and the y-axis represents the metrics.

Table 4

Evaluation of computational cost in RAE generation and reconstruction.

Attack Methods	Generation Time (s)	Reconstruction Time (s)	GPU Memory Usage (GB)
SRAE	0.51	0.48	1.85
RAE-FGSM (post)	1.24	5.86	1.44
RAE-BIM (post)	4.63	5.42	1.67
RAE-BIM (in-the-loop)	6.99	5.96	1.67
INN-RAE (Ours)	9.64	1.42	1.48

4.5. Computational resource consumption

We conducted resource consumption experiments on several reversible adversarial example generation methods compared in this paper. The evaluation was carried out from three perspectives: adversarial example generation time, reconstruction time, and peak GPU memory usage during the generation process. The experiments were performed on a platform equipped with an Intel(R) Xeon(R) Silver 4214 CPU @ 2.20GHz, 128GB of RAM, and an NVIDIA RTX 3090 GPU with 24GB of memory. The experimental results are presented in the Table 4. INN-RAE exhibits the longest generation time among the evaluated methods, taking 9.64 seconds to generate adversarial examples. This is significantly longer than SRAE (0.51 seconds), RAE-FGSM (1.24 seconds), RAE-BIM (post) (4.63 seconds), and RAE-BIM (in-the-loop) (6.99 seconds). The prolonged generation time of INN-RAE can be attributed to its complex neural network architecture, which likely involves multiple computational steps and iterative processes to ensure the quality and reversibility of the generated adversarial examples. Conversely, INN-

RAE demonstrates the short reconstruction time, recovering the original data from adversarial examples in 1.42 seconds. This is notably faster than RAE-FGSM (5.86 seconds), RAE-BIM (post) (5.42 seconds), and RAE-BIM (in-the-loop) (5.96 seconds). In RAE-BIM and RAE-FGSM, the reversible information hiding step involves computations that cannot be accelerated using GPUs. This step requires traversing every pixel of the image for reconstruction. Consequently, as the image size increases, the reconstruction speed decreases significantly. SRAE also has a relatively short reconstruction time of 0.48 seconds, but its overall performance is compromised in terms of attack success rate and perturbation quality. The rapid reconstruction capability of INN-RAE highlights its efficiency in reversible adversarial embedding, making it suitable for applications requiring quick data recovery. In terms of GPU memory consumption, INN-RAE uses 1.48 GB, which is moderate compared to other methods. SRAE consumes the most memory (1.85 GB). The memory usage of INN-RAE suggests a balance between model complexity and resource efficiency, avoiding excessive memory overhead while maintaining competitive performance. In summary, INN-RAE strikes a balance

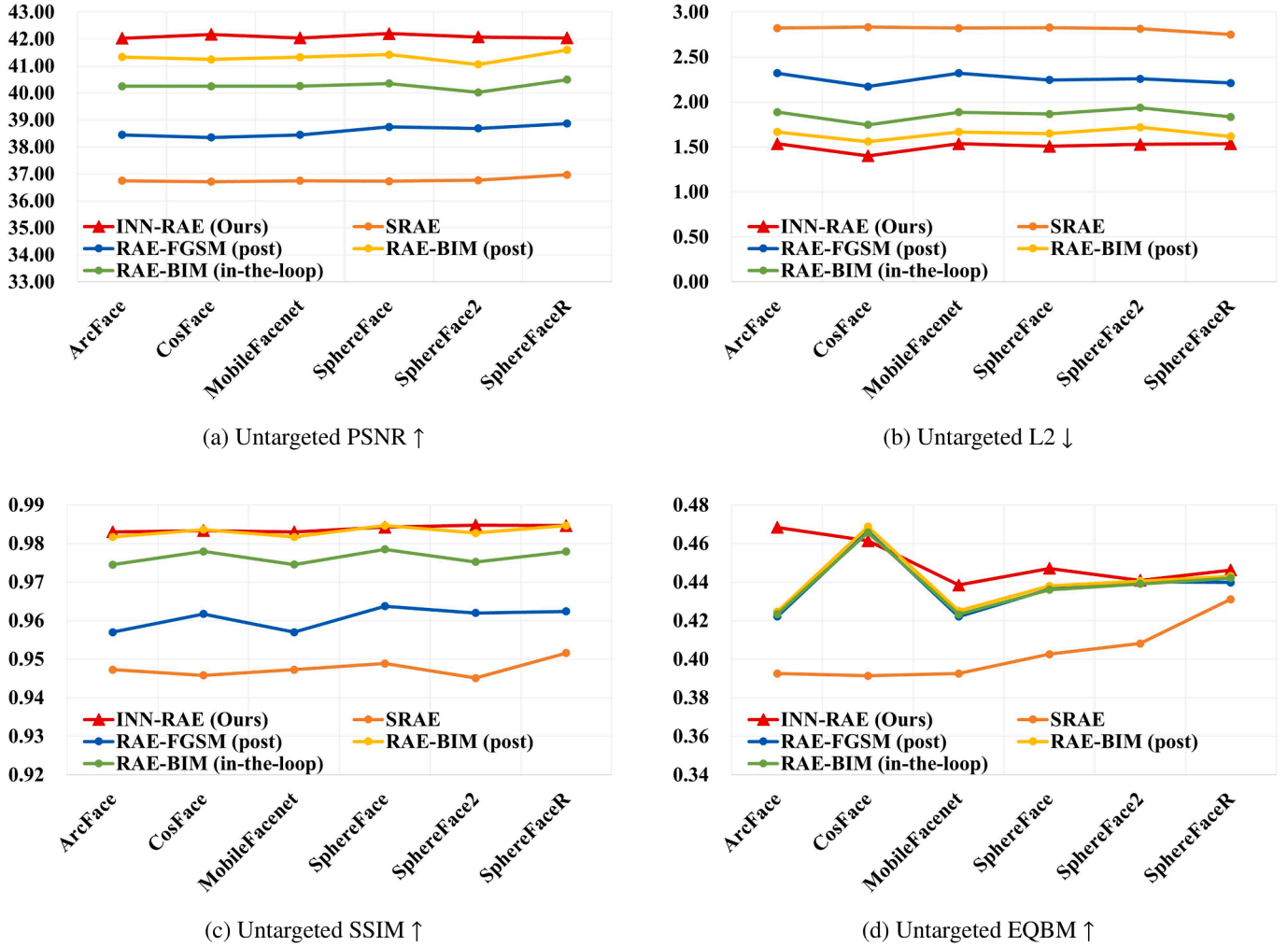


Fig. 6. Untargeted adversarial attack stealthiness performance. Where the x-axis represents the target model and the y-axis represents the metrics.

Table 5
Robustness evaluation for reversible adversarial examples.

Defense Methods	Attack Methods				
	SRAE	RAE-FGSM (post)	RAE-BIM (post)	RAE-BIM (in-the-loop)	INN-RAE (Ours)
None	0.7885	0.9035	0.9306	0.9493	0.9815
JPEG (qp = 90)	0.6710	0.4980	0.4329	0.4439	0.4581
JPEG (qp = 70)	0.5832	0.4463	0.3588	0.3713	0.4407
JPEG (qp = 50)	0.4119	0.4185	0.2901	0.3647	0.3722
Gaussian	0.7003	0.6563	0.6410	0.5968	0.6481
Super	0.7462	0.5947	0.5377	0.4943	0.5843
Random	0.6564	0.6470	0.6130	0.6044	0.6010
Average	0.6511	0.5949	0.5434	0.5464	0.5837

between memory efficiency and reconstruction speed, making it well-suited for applications prioritizing data reversibility and resource conservation. Its ability to rapidly recover original data from adversarial examples, combined with moderate memory usage, positions INN-RAE as a promising solution for security-critical applications where reversible adversarial perturbations are required. However, its high generation overhead necessitates careful consideration of application requirements and available computational resources.

4.6. Robustness evaluation

Image processing techniques developed to counteract adversarial examples and mitigate their attack potency have emerged as a validated defensive strategy. As such, when implementing adversarial example

technique, the robustness of these examples constitutes a critical consideration. We investigated the robustness of adversarial example defense methodologies and conventional image processing techniques. In alignment with the testing protocol delineated in DP-RAE (Zhu et al., 2024), we selected Random Resize and Padding (Random) (Xie et al., 2018), Gaussian Blurring (Gaussian) (Brama & Grinshpoun, 2020), JPEG Compression (JPEG) (Das et al., 2017), and Super-Resolution (Super) (Mustafa et al., 2020) as archetypal image processing algorithms to assess the robustness of adversarial example defenses. All defense parameters were rigorously preserved according to the specifications reported in the original scholarly works. The empirical findings are delineated in Table 5. The table evaluates the robustness of reversible adversarial examples generated by five attack methods (SRAE, RAE-FGSM(post), RAE-BIM(post), RAE-BIM(in-the-loop), INN-RAE) under six

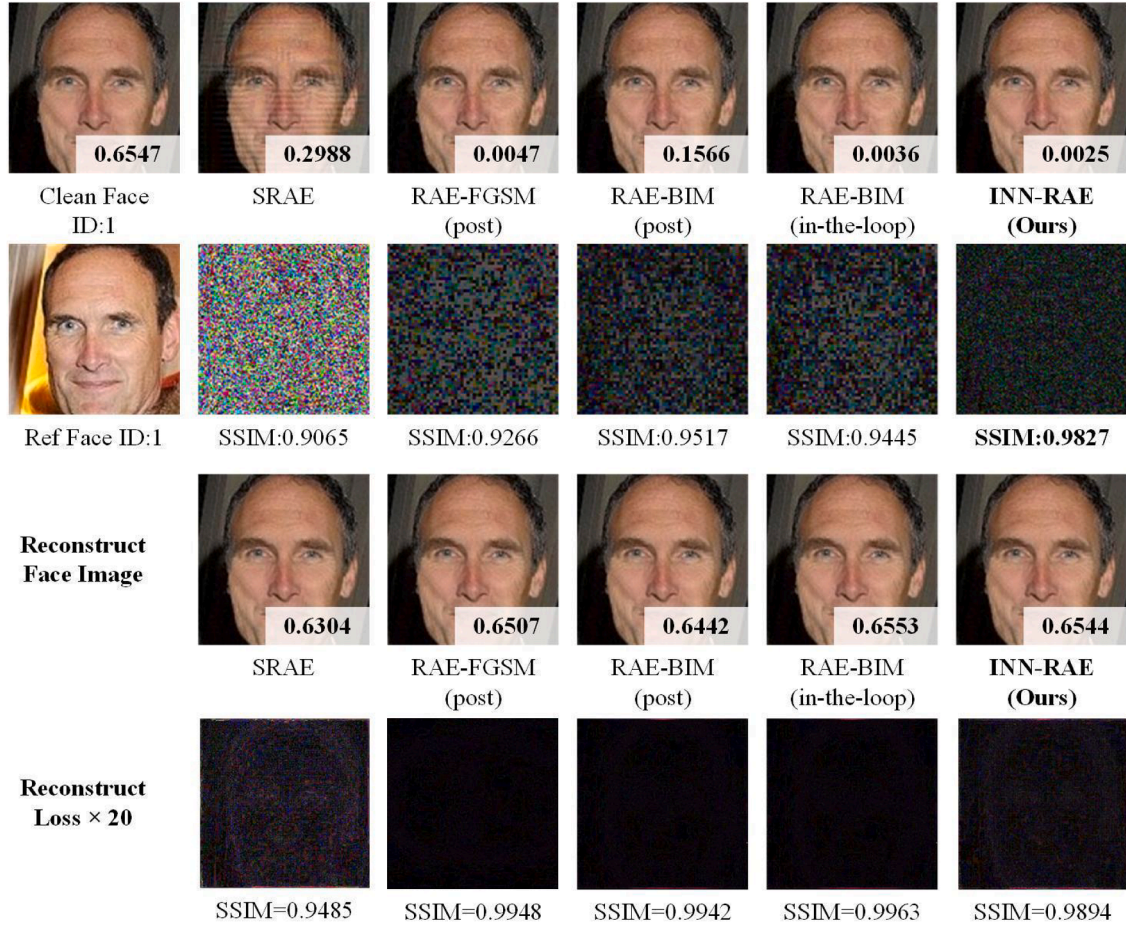


Fig. 7. Schematic diagram of **untargeted attack** on the ArcFace model using the VggFace2 dataset. The number in the lower right corner of the first row indicates the similarity between this image and the reference facial photo. The smaller the value, the better the attack effect. The second to sixth images in the second row are the corresponding added adversarial perturbations. For the convenience of observation, they have all been uniformly enlarged 20 times.

defense methods (None, JPEG compression with varying quality factors, Gaussian blurring, super-resolution, random resize/padding). Higher values indicate stronger attack efficacy after defense application. In the absence of defense ("None"), INN-RAE achieves the highest success rate (0.9815), significantly outperforming SRAE (0.7885), RAE-FGSM(post) (0.9035), and both RAE-BIM variants. This highlights INN-RAE's superior attack potency on pristine models, likely due to its invertible neural network design that enables precise perturbation tuning. **JPEG Compression:** As JPEG quality factor (qp) decreases (higher compression), INN-RAE's success rate drops from 0.4581 (qp = 90) to 0.3722 (qp = 50). While this decline is steeper than SRAE's (0.6710 $\rightarrow$ 0.4119), INN-RAE consistently outperforms RAE-BIM(post) and RAE-BIM(in-the-loop) at qp = 50 (0.3722 vs. 0.2901 and 0.3647, respectively). The invertible transformation in INN-RAE may prioritize perturbation reversibility over compression resilience, but its structured perturbation pattern still maintains higher robustness than iterative BIM methods under extreme compression. **Gaussian Blur:** INN-RAE retains a success rate of 0.6481, surpassing both RAE-BIM variants (0.6410 and 0.5968) and approaching RAE-FGSM(post) (0.6563). This suggests that INN-RAE's perturbations, which are optimized for feature-space invariance, are less disrupted by spatial smoothing. **Super-Resolution:** With a success rate of 0.5843, INN-RAE outperforms RAE-BIM(in-the-loop) (0.4943) and RAE-BIM(post) (0.5377). The invertible model's ability to preserve high-frequency features may explain its resilience against resolution enhancement, which often distorts adversarial perturbations. **Random Resize & Padding:** INN-RAE achieves 0.6010 under random resize/padding, comparable to SRAE (0.6564) and RAE-FGSM(post) (0.6470). This sta-

bility indicates that INN-RAE's perturbations are less sensitive to geometric transformations, likely due to its focus on embedding-space adversarial directions rather than pixel-level perturbations. **Average Performance:** INN-RAE's average success rate (0.5837) ranks second only to SRAE (0.6511) and surpasses all RAE-BIM variants. This demonstrates its balanced robustness across diverse defenses, combining high attack efficacy without defense with competitive resilience under practical image processing.

4.7. Adversarial transferability evaluation

The experiment evaluates the transferability of adversarial examples generated by various attack methods (SRAE, RAE-FGSM, RAE-BIM, INN-RAE) from source models (Facenet, Sphere Face, etc.) to target models (ArcFace, CosFace, Mobile, etc.) are shown in Table 6. The values represent transfer success rates, where higher values indicate stronger transferability. When ArcFace serves as the source model, the SRAE attack achieves an average transfer success rate of 0.1563, performing better on target models MobileFacenet (0.1604) and Sphere FaceR (0.1955). This indicates that adversarial examples generated by SRAE have certain transferability to specific models. The RAE-FGSM (post) method shows an average success rate of only 0.0987, particularly 0.0570 on the Sphere Face2 target model, suggesting weak transferability. Notably, our proposed INN-RAE method achieves an average of 0.1529, approaching the performance of SRAE. What sets INN-RAE apart is its superior performance on CosFace (0.1851) and MobileFacenet (0.1525). This can be attributed to its innovative integration of

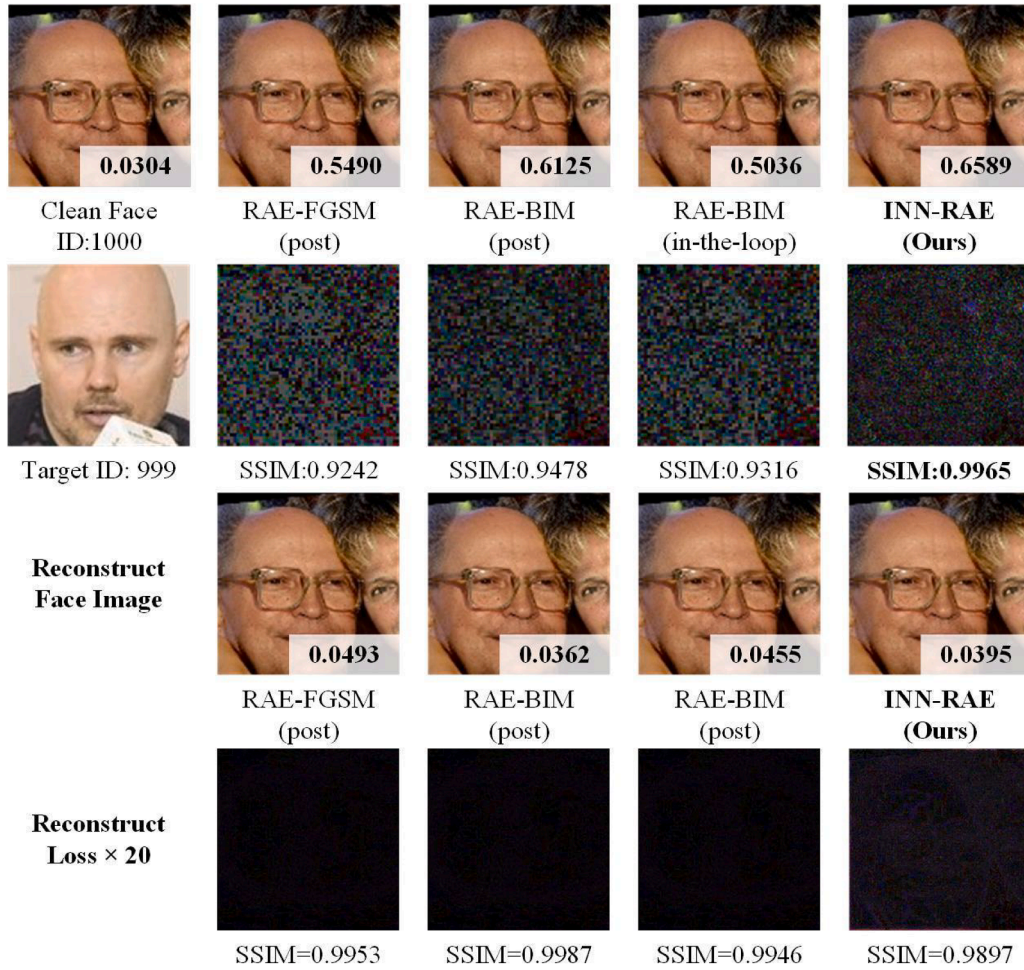


Fig. 8. Schematic diagram of **targeted attack** on the ArcFace model using the VggFace2 dataset. The number in the lower right corner of the first row indicates the similarity between this image and the target reference facial photo. The bigger the value, the better the attack effect.

Table 6
Reversible adversarial examples adversarial transferability evaluation.

Attack Methods		Target Models						
		ArcFace	CosFace	Mobile Facenet	Sphere Face	Sphere Face2	Sphere FaceR	Average
ArcFace	SRAE	–	0.1328	0.1234	0.1604	0.1694	0.1955	0.1563
	RAE-FGSM (post)	–	0.0885	0.1200	0.1140	0.0570	0.1141	0.0987
	RAE-BIM (post)	–	0.1677	0.1319	0.0823	0.1630	0.1391	0.1368
	RAE-BIM (in-the-loop)	–	0.0723	0.0914	0.1622	0.1055	0.0976	0.1058
	INN-RAE (Ours)	–	0.1325	0.1851	0.1525	0.1239	0.1706	0.1529
Sphere Face	SRAE	0.2301	0.2287	0.1993	–	0.2164	0.2001	0.2111
	RAE-FGSM (post)	0.2039	0.1169	0.2591	–	0.1911	0.2266	0.1984
	RAE-BIM (post)	0.1902	0.1758	0.1263	–	0.1450	0.1535	0.1502
	RAE-BIM (in-the-loop)	0.1253	0.2088	0.2610	–	0.1902	0.1743	0.2086
	INN-RAE (Ours)	0.2023	0.2055	0.2264	–	0.2294	0.1795	0.2102

invertible neural network architectures, which enables a more nuanced manipulation of input features while maintaining the reversibility of adversarial perturbations. By leveraging invertible transformations, INN-RAE can generate adversarial examples that better align with the decision boundaries of target models, thus enhancing transferability. When Sphere Face serves as the source model, the SRAE attack achieves an average success rate of 0.2111, significantly higher than when ArcFace is the source model. It shows strong transferability on target models ArcFace (0.2301) and CosFace (0.2287), suggesting that the feature space of Sphere Face may be closer to that of the target models. The RAE-BIM (in-the-loop) method achieves an average of 0.2086, reaching 0.2610 on the Mobile target model, exceeding SRAE's 0.1993. In this scenario,

INN-RAE also demonstrates remarkable performance with an average of 0.2102, on par with SRAE. Its effectiveness across Sphere Face2 (0.2294) and Facenet (0.1795) further validates its robustness. The success of INN-RAE can be partly explained by its ability to capture high-order feature correlations within the source model's embedding space. By modeling these complex relationships, INN-RAE generates adversarial perturbations that are more likely to be misinterpreted by target models, regardless of their architectural differences. For the ArcFace source model, RAE-BIM (post) has a higher average success rate (0.1368) than the in-the-loop strategy (0.1058), indicating that post-processing is more suitable for this source model. In contrast, for the Sphere Face source model, the in-the-loop strategy (0.2086) outperforms post-processing

Table 7

Variation of attack transferability of INN-RAE adversarial examples with adversarial attack magnitude, where the source model is ArcFace and the test dataset is CelebA.

Adversarial Magnitude	Target Models					
	CosFace	Mobile Facenet	Sphere Face	Sphere Face2	Sphere FaceR	Average
4/255	0.1325	0.1851	0.1525	0.1239	0.1706	0.1529
8/255	0.2474	0.3030	0.2478	0.2022	0.2949	0.2591
12/255	0.2661	0.3490	0.2980	0.2433	0.3433	0.2999
16/255	0.3632	0.4109	0.3436	0.2827	0.4203	0.3641
20/255	0.3375	0.4268	0.3934	0.3314	0.4526	0.3883
24/255	0.3919	0.5017	0.4326	0.3611	0.4883	0.4351
28/255	0.4220	0.4928	0.4910	0.4192	0.5451	0.4740
32/255	0.5988	0.6593	0.5549	0.4972	0.6284	0.5877

Table 8

Ablation experiments were conducted on ArcFace and VggFace2 for targeted face attacks, focusing on the four loss functions in INN-RAE.

Loss Function				Metrics								
RL_{adv}	RL_{re}	VL_{adv}	VL_{re}	ASR	$PSNR_{adv}$	$SSIM_{adv}$	$L2_{adv}$	CER	$PSNR_{re}$	$SSIM_{re}$	$L2_{re}$	RAE-P
✓				0.9523	36.8552	0.9445	1.8996	0.9054	35.9889	0.9265	5.5485	0.0788
✓	✓			0.9448	35.9868	0.9463	1.9042	0.0103	34.8756	0.8596	6.8597	0.7606
✓	✓	✓		0.8685	43.8514	0.9872	1.4972	0.0165	35.9639	0.8768	4.7258	0.7393
✓	✓	✓	✓	0.8523	42.1858	0.9765	1.5084	0.0167	39.7897	0.9784	2.0072	0.8007

(0.1502), suggesting that model architecture may affect strategy effectiveness. Compared with traditional methods, INN-RAE exhibits distinct advantages. Unlike SRAE and RAE-FGSM, which rely on straightforward perturbation generation, INN-RAE's invertible design allows for precise control over the magnitude and direction of adversarial changes. This adaptability enables INN-RAE to maintain stable transferability across diverse source and target model combinations. Additionally, while RAE-BIM methods require iterative optimization, INN-RAE achieves competitive results with a more efficient, one-step generation process, making it more practical for real-world applications. The Mobile model is vulnerable to transfer attacks under both the Sphere Face source model (RAE-BIM (in-the-loop) reaches 0.2610) and the ArcFace source model (INN-RAE reaches 0.1525), possibly due to its simpler architecture or high feature similarity to source models. CosFace shows low transfer success rates under the ArcFace source model (RAE-FGSM (post) is only 0.0885) and the Sphere Face source model (RAE-BIM (post) is only 0.1263), indicating strong robustness against adversarial examples. Notably, INN-RAE still manages to achieve relatively high transfer success rates on CosFace (0.1851 from ArcFace and 0.2055 from Sphere Face), demonstrating its superiority in overcoming the defense mechanisms of robust models. This highlights the potential of INN-RAE's invertible transformation approach in bypassing complex feature extraction and classification pipelines of target models.

Table 7 analyzes how adversarial magnitude (4/255-32/255) affects INN-RAE's attack transferability (source: ArcFace, dataset: CelebA). A clear positive correlation emerges: average transfer rate rises from 0.1529 (4/255) to 0.5877 (32/255), with accelerated growth at higher magnitudes (e.g., +0.1136 jump from 28/255 to 32/255), indicating a magnitude threshold for enhanced cross-model effectiveness. Model-specific sensitivity varies: Mobile Facenet (0.1851 $\rightarrow$ 0.6593) and SphereFaceR (0.1706 $\rightarrow$ 0.6284) are most sensitive, reflecting architectural similarity to ArcFace. SphereFace2 is most robust (0.1239 $\rightarrow$ 0.4972, lowest rates overall). These results confirm INN-RAE's transferability is magnitude-dependent, with stronger perturbations overcoming target model defenses more effectively.

4.8. Ablation study

In INN-RAE, various loss functions have been designed, including recognition and visual quality loss functions. The effects of these loss

functions were studied in ablation experiments, and target attack tests were conducted on the ArcFace model using the VggFace2 dataset. The results of the ablation experiments are shown in Table 8. As indicated in Table 8, the RL_{re} loss function slightly reduces the attack success rate of adversarial samples but significantly lowers the recognition error rate, ensuring the usability of reconstructed samples. The adversarial sample visual quality loss, VL_{adv} , significantly reduces the attack success rate but improves the visual quality of generated adversarial samples. On the other hand, the reconstructed sample quality loss, VL_{re} , slightly reduces the attack success rate and the visual quality of adversarial samples but significantly enhances the visual quality of reconstructed samples.

$$RAE - P = ASR \times SSIM_{adv} \times (1 - CER) \times SSIM_{re} \quad (23)$$

In ablation experiments, the RAE-P metric is introduced and calculated as shown in Eq. 23, to measure better the overall performance of different combinations of loss functions. Specifically, this metric not only considers the attack success rate and stealthiness of adversarial samples but also sets standards for the quality of the recovered reconstructed samples. Overall, all four loss functions effectively improve the RAE-P metric, enhancing reconstructed samples' usability and visual quality while ensuring the attack success rate and visual effects of generated adversarial samples. In our experimental evaluations, using the original image itself as the Perturbation Matrix (PM) yields better visual imperceptibility. This is because, during the forward noise injection process, the PM essentially undergoes a series of continuous transformations aimed at achieving adversarial effectiveness. Starting from the original image allows these transformations to retain more of the image's natural structure, thereby enhancing the overall imperceptibility of the resulting adversarial example. The advantage of this approach is that it reduces the additional noise interference introduced by random noise or unrelated images in the final generated output. To further validate this, we have conducted ablation studies using alternative images as the PM, including a random Gaussian noise image, and images from different identity classes. The experimental results are presented in the Table 9. The ablation experiments were conducted on ArcFace and VggFace2 for targeted face attacks.

First, in terms of ASR, all three settings achieve comparable performance (around 0.85), indicating that the choice of PM does not significantly affect the success of targeted adversarial attacks. However, while using the target identity image leads to the highest ASR (0.8615),

Table 9
Ablation experiments on PM selection.

Metrics	PM Types		
	Gaussian Noise	Target ID Face Image	Original Face Image
ASR	0.8499	0.8615	0.8523
$PSNR_{adv}$	33.9002	36.7127	42.1858
$SSIM_{adv}$	0.8989	0.9446	0.9765
$L2_{adv}$	2.4533	2.1278	1.5084
CER	0.0432	0.0558	0.0167
$PSNR_{re}$	29.1693	33.5778	39.7897
$SSIM_{re}$	0.8806	0.9074	0.9784
$L2_{re}$	3.1257	2.5006	2.0072
RAE-P	0.6437	0.6972	0.8007

this advantage comes at the cost of increased perceptual distortion and reduced reconstruction quality. In contrast, using the original face image as the PM consistently yields the best performance across multiple perceptual and reversibility metrics. Specifically, it achieves the highest $PSNR_{adv}$ (42.1858) and $SSIM_{adv}$ (0.9765), as well as the lowest $L2_{adv}$ (1.5084), indicating superior visual fidelity and minimal perturbation. This is because using the original image as PM allows the noise injection process to introduce adversarial perturbations in a structure-preserving manner, thereby avoiding unnecessary distortion. On the other hand, Gaussian noise as PM introduces substantial perturbations, leading to degraded PSNR and SSIM. In terms of semantic preservation, the original image also results in the lowest CER (0.0167), compared to 0.0432 for Gaussian noise and 0.0558 for the target identity image. This suggests that using unrelated images as PMs may introduce semantic inconsistencies or identity blending effects. Furthermore, the reconstruction quality from the adversarial image back to the original image is also highest when the original image is used as PM. It achieves the best $PSNR_{re}$ (39.7897), $SSIM_{re}$ (0.9784), and $L2_{re}$ (2.0072), demonstrating that reversibility is better maintained when the starting point is closely aligned with the target structure. Lastly, the RAE-P metric confirms that identity consistency is best preserved using the original PM (RAE-P = 0.8007), while Gaussian noise yields the lowest value (0.6437). This underscores the importance of PM similarity in maintaining representation integrity in the embedding space. In summary, while all three PM types enable successful adversarial attacks, only the original image PM simultaneously ensures high imperceptibility, semantic consistency, and reversibility. These findings justify our design choice and highlight the crucial role of PM selection in reversible adversarial sample generation.

5. Conclusion

This paper focuses on the issue of image privacy protection in social media scenarios. It proposes a high visual quality reversible adversarial example attack method based on invertible neural networks INN-RAE. This method generates adversarial examples with high visual concealment through invertible neural networks. These adversarial examples can prevent unauthorized AI from extracting private information using samples. When authorized AI attempts to use these samples, the same invertible neural network can be used to restore and correctly identify them. In both image classification and face recognition tasks, INN-RAE achieves high attack success rates and visual quality across different datasets and target networks of varying structures while also restoring high-quality samples. In future work, we will consider migrating this method to black-box attacks and improving the robustness of reversible adversarial examples.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- Aneja, S., Markhasin, L., & Nießner, M. (2022). TAFIM: targeted adversarial attacks against facial image manipulations. In S. Avidan, G. J. Brostow, M. Cissé, G. M. Fariella, & T. Hassner (Eds.), *Computer vision - ECCV 2022 - 17th european conference, tel aviv, israel, october 23-27, 2022, proceedings, part XIV* (pp. 58–75). Springer (vol. 13674). Lecture Notes in Computer Science.
- Ardizzone, L., Kruse, J., Lüth, C. T., Bracher, N., Rother, C., & Köthe, U. (2020). Conditional invertible neural networks for diverse image-to-image translation. In Z. Akata, A. Geiger, & T. Sattler (Eds.), *Pattern recognition - 42nd DAGM german conference, DAGM GCPR 2020, tübingen, germany, september 28 - october 1, 2020, proceedings* (pp. 373–387). Springer (vol. 12544). Lecture Notes in Computer Science.
- Ardizzone, L., Lüth, C. T., Kruse, J., Rother, C., & Köthe, U. (2019). Guided image generation with conditional invertible neural networks. *CoRR, abs/1907.02392*. (pp. 1–11).
- Brama, H., & Grinshpoun, T. (2020). Heat and blur: An effective and fast defense against adversarial examples. *CoRR, abs/2003.07573*. (pp. 1–7).
- Carlini, N., & Wagner, D. A. (2017). Towards evaluating the robustness of neural networks. In *2017 IEEE symposium on security and privacy* (pp. 39–57). IEEE Computer Society.
- Chen, S., Liu, Y., Gao, X., & Han, Z. (2018). Mobilefacenet: Efficient CNNs for accurate real-time face verification on mobile devices. In J. Zhou, Y. Wang, Z. Sun, Z. Jia, J. Feng, S. Shan, K. Ubul, & Z. Guo (Eds.), *Biometric recognition - 13th chinese conference, CCBR 2018, urumqi, china, august 11-12, 2018, proceedings* (pp. 428–438). Springer (vol. 10996). Lecture Notes in Computer Science.
- Chen, Z., Wang, Z., Huang, J., Zhao, W., Liu, X., & Guan, D. (2023). Imperceptible adversarial attack via invertible neural networks. In B. Williams, Y. Chen, & J. Neville (Eds.), *Thirty-seventh AAAI conference on artificial intelligence, AAAI 2023, thirty-fifth conference on innovative applications of artificial intelligence, IAAI 2023, thirteenth symposium on educational advances in artificial intelligence, EAAI 2023, washington, dc, usa, february 7-14, 2023* (pp. 414–424). AAAI Press.
- Chin-Yuan, Y., Hsi-Wen, C., Hong-Han, S., De-Nian, Y., & Ming-Syan, C. (2021). Attack as the best defense: Nullifying image-to-image translation GANs via limit-aware adversarial attack. In *Proceedings of the IEEE/CVF international conference on computer vision (ICCV)* (pp. 16188–16197).
- Chin-Yuan, Y., Hsi-Wen, C., Shang-Lun, T., & Sheng-De, W. (2020). Disrupting image-translation-based deepfake algorithms with adversarial attacks. In *Proceedings of the IEEE/CVF winter conference on applications of computer vision (WACV) workshops*.
- Chong, M., Youmin, X., Jiechong, S., Chen, Z., Bernard, G., & Jian, Z. (2023). Large-capacity and flexible video steganography via invertible neural network. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)* (pp. 22606–22615).
- Das, N., Shanbhogue, M., Chen, S., Hohman, F., Chen, L., Kounavis, M. E., & Chau, D. H. (2017). Keeping the bad guys out: Protecting and vaccinating deep learning with JPEG compression. *CoRR, abs/1705.02900*.
- Dinh, L., Krueger, D., & Bengio, Y. (2015). NICE: non-linear independent components estimation. In Y. Bengio, & Y. LeCun (Eds.), *3rd international conference on learning representations, ICLR 2015, san diego, ca, usa, may 7-9, 2015, workshop track proceedings*.
- Dinh, L., Sohl-Dickstein, J., & Bengio, S. (2017). Density estimation using real NVP. In *5th international conference on learning representations, ICLR 2017, toulon, france, april 24-26, 2017, conference track proceedings*. OpenReview.net.
- Dong, Y., Liao, F., Pang, T., Su, H., Zhu, J., Hu, X., & Li, J. (2018). Boosting adversarial attacks with momentum. In *2018 IEEE conference on computer vision and pattern recognition*, (pp. 9185–9193). Computer Vision Foundation / IEEE Computer Society.
- Gao, H., Zhuang, L., van der, M. L., & Weinberger Kilian, Q. (2017). Densely connected convolutional networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition (CVPR)*.
- Goodfellow, I. J., Shlens, J., & Szegedy, C. (2015). Explaining and harnessing adversarial examples. In Y. Bengio, & Y. LeCun (Eds.), *3rd international conference on learning representations*.
- Hao, W., Yitong, W., Zheng, Z., Xing, J., Dihong, G., Jingchao, Z., Zhifeng, L., & Wei, L. (2018). Cosface: Large margin cosine loss for deep face recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition (CVPR)*.
- Jiankang, D., Jia, G., Niannan, X., & Stefanos, Z. (2019). Arcface: Additive angular margin loss for deep face recognition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)*.
- Jun-Jie, H., & Luigi, D. P. (2022). Winnet: Wavelet-inspired invertible network for image denoising. *IEEE Transactions on Image Processing*, 31, 4377–4392.
- Liu, J., Zhang, W., Fukuchi, K., Akimoto, Y., & Sakuma, J. (2023). Unauthorized AI cannot recognize me: Reversible adversarial example. *Pattern Recognit.*, 134, 109048.
- Madry, A., Makelov, A., Schmidt, L., Tsipras, D., & Vladu, A. (2018). Towards deep learning models resistant to adversarial attacks. In *6th international conference on learning representations*.
- Mustafa, A., Khan, S. H., Hayat, M., Shen, J., & Shao, L. (2020). Image super-resolution as a defense against adversarial attacks. *IEEE Trans. Image Process.*, 29, 1711–1724.
- Ou, F., Chen, B., Li, C., Wang, S., & Kwong, S. (2023). Troubleshotting ethnic quality bias with curriculum domain adaptation for face image quality assessment. In *IEEE/CVF international conference on computer vision, ICCV 2023, paris, france, october 1-6, 2023* (pp. 20661–20672). IEEE.

- Qiong, C., Li, S., Weidi, X., Parkhi Omkar, M., & Andrew, Z. (2018). Vggface2: A dataset for recognising faces across pose and age. In *2018 13th IEEE international conference on automatic face and gesture recognition (FG 2018)* (pp. 67–74).
- Rabhi, M., Bakiras, S., & Pietro, R. D. (2024). Audio-deepfake detection: Adversarial attacks and countermeasures. *Expert Syst. Appl.*, 250, 123941.
- Ran, Y., Zhang, A., Li, M., Tang, W., & Wang, Y. (2025). Black-box adversarial attacks against image quality assessment models. *Expert Syst. Appl.*, 260, 125415.
- Rongke, L., Dong, W., Yizhi, R., Zhen, W., Kaitian, G., Qianqian, Q., & Xiaolei, L. (2024). Unstoppable attack: Label-only model inversion via conditional diffusion model. *IEEE Transactions on Information Forensics and Security*, 19, 3958–3973.
- Ruiz, N., Bargal, S. A., & Sclaroff, S. (2020). Disrupting deepfakes: Adversarial attacks against conditional image translation networks and facial manipulation systems. In A. Bartoli, & A. Fusiello (Eds.), *Computer vision - ECCV 2020 workshops - glasgow, uk, august 23-28, 2020, proceedings, part IV* (pp. 236–251). Springer (vol. 12538). Lecture Notes in Computer Science.
- Shan, S., Wenger, E., Zhang, J., Li, H., Zheng, H., & Zhao, B. Y. (2020). Fawkes: Protecting privacy against unauthorized deep learning models. In *29th USENIX security symposium (USENIX security 20)* (pp. 1589–1604). USENIX Association.
- Shao-Ping, L., Rong, W., Tao, Z., & Rosin Paul, L. (2021). Large-capacity image steganography based on invertible neural networks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)* (pp. 10816–10825).
- Shengshan, H., Xiaogeng, L., Yechao, Z., Minghui, L., Yu, Z. L., Hai, J., & Libing, W. (2022). Protecting facial privacy: Generating adversarial identity masks via style-robust makeup transfer. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)* (pp. 15014–15023).
- Sutanu, B., & Kumar, B. P. (2023). Self supervised low dose computed tomography image denoising using invertible network exploiting inter slice congruence. In *Proceedings of the IEEE/CVF winter conference on applications of computer vision (WACV)* (pp. 5614–5623).
- Tang, L., Ye, D., Lv, Y., Chen, C., & Zhang, Y. (2024). Once and for all: Universal transferable adversarial perturbation against deep hashing-based facial image retrieval. In M. J. Wooldridge, J. G. Dy, & S. Natarajan (Eds.), *Thirty-eighth AAAI conference on artificial intelligence, AAAI 2024* (pp. 5136–5144). AAAI Press.
- Thanh, V. L., Hao, P., Hoang, N. T., Quan, D., Tran Ngoc, N., & Anh, T. (2023). Anti-dreambooth: Protecting users from personalized text-to-image synthesis. In *Proceedings of the IEEE/CVF international conference on computer vision (ICCV)* (pp. 2116–2127).
- Wang, R., Huang, Z., Chen, Z., Liu, L., Chen, J., & Wang, L. (2022). Anti-forgery: Towards a stealthy and robust deepfake disruption attack via adversarial perceptual-aware perturbations. In L. De Raedt (Ed.), *Proceedings of the thirty-first international joint conference on artificial intelligence, IJCAI 2022, vienna, austria, 23-29 july 2022* (pp. 761–767). ijcai.org.
- Wei, X., Yan, H., & Li, B. (2022). Sparse black-box video attack with reinforcement learning. *International Journal of Computer Vision*, 130(6), 1459–1473.
- Wei, X., Zhu, J., Yuan, S., & Su, H. (2019). Sparse adversarial perturbations for videos. In *The thirty-third AAAI conference on artificial intelligence* (pp. 8973–8980). AAAI Press.
- Wei, Z., Chen, Z., Wu, Z., Jiang, Y., (2022a). Boosting the transferability of video adversarial examples via temporal translation. In *Thirty-sixth AAAI conference on artificial intelligence* (pp. 2659–2667). AAAI Press.
- Wei, Z., Chen, J., Wu, Z., Jiang, Y., (2022b). Cross-modal transferable adversarial attacks from images to videos. In *IEEE/CVF conference on computer vision and pattern recognition* (pp. 15044–15053). IEEE.
- Weiyang, L., Yandong, W., Zhiding, Y., Ming, L., Bhiksha, R., & Le, S. (2017). Sphereface: Deep hypersphere embedding for face recognition. In *The IEEE conference on computer vision and pattern recognition (CVPR)*.
- Xie, C., Wang, J., Zhang, Z., Ren, Z., & Yuille, A. L. (2018). Mitigating adversarial effects through randomization. In *6th international conference on learning representations, ICLR 2018, vancouver, bc, canada, april 30 - may 3, 2018, conference track proceedings*. OpenReview.net.
- Xingxing, W., Songping, W., & Huanqian, Y. (2023). Efficient robustness assessment via adversarial spatial-temporal focus on videos. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(9), 10898–10912.
- Xiong, L., Wu, Y., Yu, P., & Zheng, Y. (2023). A black-box reversible adversarial example for authorizable recognition to shared images. *Pattern Recognit.*, 140, 109549.
- Ye, T., Jingwen, P., Shangshang, Y., Xingyi, Z., Shuping, H., & Yaochu, J. (2023). Imperceptible and sparse adversarial attacks via a dual-population-based constrained evolutionary algorithm. *IEEE Transactions on Artificial Intelligence*, 4(2), 268–281.
- Yin, Z., Chen, L., Lyu, W., & Luo, B. (2023). Reversible attack based on adversarial perturbation and reversible data hiding in YUV colorspace. *Pattern Recognition Letters*, 166, 1–7.
- Youmin, X., Chong, M., Yujie, H., Jingfen, X., & Jian, Z. (2022). Robust invertible image steganography. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)* (pp. 7875–7884).
- Yuan, L., Liu, L., Pu, X., Li, Z., Li, H., & Gao, X. (2022). Pro-face: A generic framework for privacy-preserving recognizable obfuscation of face images. In J. Magalhães, A. D. Bimbo, S. Satoh, N. Sebe, X. Alameda-Pineda, Q. Jin, V. Oria, & L. Toni (Eds.), *MM '22: The 30th ACM international conference on multimedia, lisboa, portugal, october 10 - 14, 2022* (pp. 1661–1669). ACM.
- Yunqian, W., Bo, L., Jingyi, C., Rong, X., & Li, S. (2023). Divide and conquer: a two-step method for high quality face de-identification with model explainability. In *Proceedings of the IEEE/CVF international conference on computer vision (ICCV)* (pp. 5148–5157).
- Zhang, J., Wang, J., Hao, W., & Xiangyang, L. (2023). Self-recoverable adversarial examples: A new effective protection mechanism in social networks. *IEEE Transactions on Circuits and Systems for Video Technology*, 33(2), 562–574.
- Zhang, Y., Yin, Z., Li, Y., Yin, G., Yan, J., Shao, J., & Liu, Z. (2020). Celeba-spoof: Large-scale face anti-spoofing dataset with rich annotations. In A. Vedaldi, H. Bischof, T. Brox, & J. Frahm (Eds.), *Computer vision - ECCV 2020 - 16th european conference, glasgow, uk, august 23-28, 2020, proceedings, part XII* (pp. 70–85). Springer (vol. 12357). Lecture Notes in Computer Science.
- Zhao, M., Bo, W., Wei, W., Yuqiu, K., Tianhang, Z., & Kui, R. (2022a). Guided erasable adversarial attack (GEAA) toward shared data protection. *IEEE Transactions on Information Forensics and Security*, 17, 2468–2482.
- Zhao, Z., Xu, K., Jiang, X., & Sun, T. (2022b). Research on video adversarial attack with long living cycle. In J. Cussens, & K. Zhang (Eds.), *Uncertainty in artificial intelligence, proceedings of the thirty-eighth conference on uncertainty in artificial intelligence* (pp. 2374–2382). PMLR (vol. 180). Proceedings of Machine Learning Research.
- Zhen, C., Xiuli, C., Zhihua, G., Binjie, W., & Yushu, Z. (2024). Rae-vwp: A reversible adversarial example-based privacy and copyright protection method of medical images for internet of medical things. *IEEE Internet of Things Journal*, 11(11), 20757–20768.
- Zhu, J., Du, X., Zhou, J., Pun, C., Xu, Q., & Liu, X. (2024). DP-RAE: A dual-phase merging reversible adversarial example for image privacy protection. In J. Cai, M. S. Kankanhalli, B. Prabhakaran, S. Boll, R. Subramanian, L. Zheng, V. K. Singh, P. César, L. Xie, & D. Xu (Eds.), *Proceedings of the 32nd ACM international conference on multimedia, MM 2024, melbourne, vic, australia, 28 october 2024 - 1 november 2024* (pp. 671–680). ACM.